
Lab 4 – Static Timing Analysis

Lab 4 – COMPENG 4DV4

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12*

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```
1
pt_shell> report_analysis_coverage
*****
Report : analysis_coverage
Design : ALU
Version: U-2022.12
Date   : Tue Mar 10 01:08:23 2026
*****
```

Type of Check	Total	Met	Violated	Untested
setup	224	0 (0%)	56 (25%)	168 (75%)
hold	224	30 (13%)	26 (12%)	168 (75%)
recovery	112	56 (50%)	0 (0%)	56 (50%)
removal	112	0 (0%)	56 (50%)	56 (50%)
min_pulse_width	224	0 (0%)	112 (50%)	112 (50%)
out_setup	8	0 (0%)	8 (100%)	0 (0%)
out_hold	8	8 (100%)	0 (0%)	0 (0%)
All Checks	912	94 (10%)	258 (28%)	560 (61%)

7

6 Basic Reports:

```
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```

INVx2_ASAP7_75t_R	asap7sc7p5t_INVBUF_RVT_TT_nldm_211120	0.06	12	0.70
INVx3_ASAP7_75t_R	asap7sc7p5t_INVBUF_RVT_TT_nldm_211120	0.07	1	0.07
INVx4_ASAP7_75t_R	asap7sc7p5t_INVBUF_RVT_TT_nldm_211120	0.09	3	0.26
INVx5_ASAP7_75t_R	asap7sc7p5t_INVBUF_RVT_TT_nldm_211120	0.10	2	0.20
MAND2x1p5_ASAP7_75t_R	asap7sc7p5t_SIMPLE_RVT_TT_nldm_211120	0.12	6	0.70
MAND2xp5_ASAP7_75t_R	asap7sc7p5t_SIMPLE_RVT_TT_nldm_211120	0.06	7	0.41
MAND3xp33_ASAP7_75t_R	asap7sc7p5t_SIMPLE_RVT_TT_nldm_211120	0.07	6	0.44
MOR2x1_ASAP7_75t_R	asap7sc7p5t_SIMPLE_RVT_TT_nldm_211120	0.09	6	0.52
MOR2x1p5_ASAP7_75t_R	asap7sc7p5t_SIMPLE_RVT_TT_nldm_211120	0.12	5	0.58
MOR2x2_ASAP7_75t_R	asap7sc7p5t_SIMPLE_RVT_TT_nldm_211120	0.09	2	0.17
MOR2x6_ASAP7_75t_R	asap7sc7p5t_SIMPLE_RVT_TT_nldm_211120	0.17	1	0.17
MOR3x1_ASAP7_75t_R	asap7sc7p5t_SIMPLE_RVT_TT_nldm_211120	0.09	2	0.17
PIEH1x1_ASAP7_75t_R	asap7sc7p5t_SIMPLE_RVT_TT_nldm_211120	0.04	1	0.04
PIELOx1_ASAP7_75t_R	asap7sc7p5t_SIMPLE_RVT_TT_nldm_211120	0.04	1	0.04
MOR2x2_ASAP7_75t_R	asap7sc7p5t_SIMPLE_RVT_TT_nldm_211120	0.10	3	0.48
Total 37 references				53.83

Figure 1: Reference report

Output for the **report_reference** is shown above. This report indicates the design uses 37 reference cells with an area of 53.83.

7 Timing Analysis

```

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File Edit View Search Terminal Help
*****
Startpoint: reg_B_reg[3]
(rising edge-triggered flip-flop clocked by clk)
Endpoint: alu_out_reg[7]
(rising edge-triggered flip-flop clocked by clk)
Last common pin: clk
Path Group: clk
Path Type: max
Point          Incr      Path
-----
clock clk (rise edge)          0.00      0.00
clock network delay (propagated) 0.00      0.00
reg_B_reg[3]/CLK (DFFASRHQNX1_ASAP7_75t_R) 0.00      0.00 r
reg_B_reg[3]/Q0 (DFFASRHQNX1_ASAP7_75t_R) 49.87     49.87 f
U157/Y (BUFx5_ASAP7_75t_R)     27.28     77.15 f
U156/Y (BUFx16f_ASAP7_75t_R)   22.51     99.66 f
sub_43/U4/Y (INNx4_ASAP7_75t_R) 6.93     106.59 r
sub_43/U39/Y (NOR2x1_ASAP7_75t_R) 6.96     113.55 f
sub_43/U30/Y (AO221x2_ASAP7_75t_R) 37.20     150.75 f
sub_43/U29/Y (AO131xp3_ASAP7_75t_R) 20.15     170.90 r
sub_43/U26/Y (BUFx2_ASAP7_75t_R) 22.71     193.61 r
sub_43/U27/Y (AO221x2_ASAP7_75t_R) 31.83     225.45 r
sub_43/U28/Y (INNx2_ASAP7_75t_R) 7.32     232.76 f
sub_43/U35/Y (XOR2x2_ASAP7_75t_R) 24.70     257.47 f
U242/Y (AND3x1_ASAP7_75t_R)    14.42     271.89 f
U190/Y (AO221x2_ASAP7_75t_R)  34.60     306.49 f
U191/Y (INNx1_ASAP7_75t_R)     8.50     314.99 r
alu_out_reg[7]/D (DFFASRHQNX1_ASAP7_75t_R) 0.00     314.39 r
data arrival time                314.39

clock clk (rise edge)          10.00     10.00
clock network delay (propagated) 0.00     10.00
clock reconvergence pessimism   0.00     10.00
clock uncertainty                -0.50      9.50
alu_out_reg[7]/CLK (DFFASRHQNX1_ASAP7_75t_R) 9.50      9.50 r
library setup time             -18.85     -9.35
data required time              -9.35     -9.35
-----
data required time              -9.35
data arrival time              -314.39
-----
slack (VIOLATED)                -323.73

```

Figure 2: Timing analysis report

The timing report shows that the design does not meet timing for a clock period of 10ps (100GHz). The data required time is -9.35 ps, meaning data must reach flip flop inputs 9.35ps before the beginning of a clock cycle. The actual data arrival time is 314.39 ps, causing a slack violation of 323.73ps. The critical path is the path between reg_B_reg[3] and alu_out_reg[7].

8 Modify Clock Period:

```

Startpoint: reg_B_reg[3]
(rising edge-triggered flip-flop clocked by clk)
Endpoint: alu_out_reg[7]
(rising edge-triggered flip-flop clocked by clk)
Last common pin: clk
Path Group: clk
Path Type: max

```

Point	Incr	Path
clock clk (rise edge)	0.00	0.00
clock network delay (propagated)	0.00	0.00
reg_B_reg[3]/CLK (DFFASRHQNx1_ASAP7_75t_R)	0.00	0.00 r
reg_B_reg[3]/QN (DFFASRHQNx1_ASAP7_75t_R)	49.87	49.87 f
U157/Y (BUFx5_ASAP7_75t_R)	27.28	77.15 f
U156/Y (BUFx16f_ASAP7_75t_R)	22.51	99.66 f
sub_43/U4/Y (INVx4_ASAP7_75t_R)	6.93	106.59 r
sub_43/U39/Y (NOR2x1_ASAP7_75t_R)	6.96	113.55 f
sub_43/U38/Y (A0221x2_ASAP7_75t_R)	37.20	150.75 f
sub_43/U29/Y (A0131xp33_ASAP7_75t_R)	20.15	170.90 r
sub_43/U26/Y (BUFx2_ASAP7_75t_R)	22.71	193.61 r
sub_43/U27/Y (A0221x2_ASAP7_75t_R)	31.83	225.45 r
sub_43/U28/Y (INVx2_ASAP7_75t_R)	7.32	232.76 f
sub_43/U35/Y (XOR2x2_ASAP7_75t_R)	24.70	257.47 f
U242/Y (AND3x1_ASAP7_75t_R)	14.42	271.89 f
U190/Y (A0221x2_ASAP7_75t_R)	34.00	305.89 f
U191/Y (INVx1_ASAP7_75t_R)	8.50	314.39 r
alu_out_reg[7]/D (DFFASRHQNx1_ASAP7_75t_R)	0.00	314.39 r
data arrival time		314.39
clock clk (rise edge)	2.00	2.00
clock network delay (propagated)	0.00	2.00
clock reconvergence pessimism	0.00	2.00
clock uncertainty	-0.50	1.50
alu_out_reg[7]/CLK (DFFASRHQNx1_ASAP7_75t_R)		1.50 r
library setup time	-18.85	-17.35
data required time		-17.35
data required time		-17.35
data arrival time		-314.39
slack (VIOLATED)		-331.73

Figure 3: Timing analysis report with new clock period 2ps

The new timing report produced for the 2ps (500 GHz) clock can be seen above. Because the clock period has been decreased, timing still does not pass, with a slack violation of 331.73ps, worse than in part 7. The timing failure is due to a setup violation. The line **Path Type: max** indicates that PrimeTime checks the maximum amount of time for signals to propagate through the critical path. This maximum exceeded the data required time, and a setup violation occurred.

To determine the maximum clock frequency, we must determine the required clock period length. This is equal to the sum of data arrival time, setup time, and uncertainty. This is equal to:

$$T_{min} = 314.39ps + 18.85ps + 0.50ps = 333.74ps$$

This $T_{\min}$ corresponds to a clock frequency of 2.996 GHz. When the clock period is adjusted to 333.98ps, the timing report shows that timing is met.

```

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File Edit View Search Terminal Help
reg_B_reg[3]/CLK (DFFASRHQNx1_ASAP7_75t_R)      0.00      0.00 r
reg_B_reg[3]/QN (DFFASRHQNx1_ASAP7_75t_R)      49.87     49.87 f
U157/Y (BUFx5_ASAP7_75t_R)                      27.28     77.15 f
U156/Y (BUFx16f_ASAP7_75t_R)                   22.51     99.66 f
sub_43/U4/Y (INVx4_ASAP7_75t_R)                 6.93     106.59 r
sub_43/U39/Y (NOR2x1_ASAP7_75t_R)               6.96     113.55 f
sub_43/U30/Y (AO221x2_ASAP7_75t_R)             37.20     150.75 f
sub_43/U29/Y (AOI31xp33_ASAP7_75t_R)           20.15     170.90 r
sub_43/U26/Y (BUFx2_ASAP7_75t_R)               22.71     193.61 r
sub_43/U27/Y (AO221x2_ASAP7_75t_R)             31.83     225.45 r
sub_43/U28/Y (INVx2_ASAP7_75t_R)                7.32     232.76 f
sub_43/U35/Y (XOR2x2_ASAP7_75t_R)              24.70     257.47 f
U242/Y (AND3x1_ASAP7_75t_R)                    14.42     271.89 f
U190/Y (AO221x2_ASAP7_75t_R)                   34.00     305.89 f
U191/Y (INVx1_ASAP7_75t_R)                      8.50     314.39 r
alu_out_reg[7]/D (DFFASRHQNx1_ASAP7_75t_R)      0.00     314.39 r
data arrival time                               314.39

clock clk (rise edge)                          333.80     333.80
clock network delay (propagated)                 0.00     333.80
clock reconvergence pessimism                    0.00     333.80
clock uncertainty                               -0.50     333.30
alu_out_reg[7]/CLK (DFFASRHQNx1_ASAP7_75t_R)    333.30     333.30 r
library setup time                             -18.85     314.45
data required time                              314.45

-----
data required time                              314.45
data arrival time                              -314.39
-----
slack (MET)                                     0.07

1
Information: Defining new variable 'design_clock'. (CMD-041)
1
pt_shell>

```

Figure 4: Timing analysis report with $T_{\min}$ clock period 333.8ps

9 Restore Clock Period:

```

Startpoint: reg_B_reg[3]
(rising edge-triggered flip-flop clocked by clk)
Endpoint: alu_out_reg[7]
(rising edge-triggered flip-flop clocked by clk)
Last common pin: clk
Path Group: clk
Path Type: max

```

Point	Incr	Path
clock clk (rise edge)	0.00	0.00
clock network delay (propagated)	0.00	0.00
reg_B_reg[3]/CLK (DFFASRHQNX1 ASAP7_75t_R)	0.00	0.00 r
reg_B_reg[3]/QN (DFFASRHQNX1 ASAP7_75t_R)	49.87	49.87 f
U157/Y (BUFx5 ASAP7_75t_R)	27.28	77.15 f
U156/Y (BUFx16f ASAP7_75t_R)	22.51	99.66 f
sub_43/U4/Y (INVx4 ASAP7_75t_R)	6.93	106.59 r
sub_43/U39/Y (NORx1 ASAP7_75t_R)	6.96	113.55 f
sub_43/U30/Y (AO221x2 ASAP7_75t_R)	37.20	150.75 f
sub_43/U29/Y (AOI31xp33 ASAP7_75t_R)	20.15	170.90 r
sub_43/U26/Y (BURx2 ASAP7_75t_R)	22.71	193.61 r
sub_43/U27/Y (AO221x2 ASAP7_75t_R)	31.83	225.45 r
sub_43/U28/Y (INVx2 ASAP7_75t_R)	7.32	232.76 f
sub_43/U35/Y (XORx2 ASAP7_75t_R)	24.70	257.47 f
U242/Y (AND3x1 ASAP7_75t_R)	14.42	271.89 f
U190/Y (AO221x2 ASAP7_75t_R)	34.00	305.89 f
U191/Y (INVx1 ASAP7_75t_R)	8.50	314.39 r
alu_out_reg[7]/D (DFFASRHQNX1 ASAP7_75t_R)	0.00	314.39 r
data arrival time		314.39
clock clk (rise edge)	10.00	10.00
clock network delay (propagated)	0.00	10.00
clock reconvergence pessimism	0.00	10.00
clock uncertainty	-0.50	9.50
alu_out_reg[7]/CLK (DFFASRHQNX1 ASAP7_75t_R)		9.50 r
library setup time	-18.85	-9.35
data required time		-9.35
data arrival time		-314.39
slack (VIOLATED)		-323.73

Figure 5: Timing analysis report with clock period restored to 10ps

With the clock period restored to 10ps, the design fails timing by the same amount as in Part 7, and with the same critical path. Because the design wasn't changed, the timing analysis results were unchanged.

10 Restore Clock Period:

```

v72180@vcl-vm0-167:Lab4
File Edit View Search Terminal Help
Warning: Nothing implicitly matched all violations (SEL-003)
Error: Nothing matched for object_list (SEL-005)
0
pt_shell> report analysis coverage
*****
Report : analysis_coverage
Design : ALU
Version: U-2022.12
Date   : Tue Mar 10 00:44:12 2026
*****

```

Type of Check	Total	Met	Violated	Untested
setup	224	0 (0%)	56 (25%)	168 (75%)
hold	224	30 (13%)	26 (12%)	168 (75%)
recovery	112	56 (50%)	0 (0%)	56 (50%)
removal	112	0 (0%)	56 (50%)	56 (50%)
min_pulse_width	224	0 (0%)	112 (50%)	112 (50%)
out_setup	8	0 (0%)	8 (100%)	0 (0%)
out_hold	8	8 (100%)	0 (0%)	0 (0%)
All checks	912	94 (10%)	258 (28%)	560 (61%)

Figure 6: Constraint analysis report

Constraint analysis shows 56 setup, 26 hold, 56 removal, 112 min_pulse_width, and 8 out_setup violations, for a total of 258 violations.

Hold violations occur when data reaches the input of a flip flop too quickly after a positive clock edge, essentially overwriting the data that is on the Q side. Common methods of fixing hold violations are the use of buffers to increase propagation delay of the signal causing the hold violation.

11 Final Checkpoint:

```
1
pt_shell> report_analysis_coverage
*****
Report : analysis_coverage
Design : ALU
Version: U-2022.12
Date   : Tue Mar 10 01:08:23 2026
*****
```

Type of Check	Total	Met	Violated	Untested
-----	-----	-----	-----	-----
setup	224	0 (0%)	56 (25%)	168 (75%)
hold	224	30 (13%)	26 (12%)	168 (75%)
recovery	112	56 (50%)	0 (0%)	56 (50%)
removal	112	0 (0%)	56 (50%)	56 (50%)
min_pulse_width	224	0 (0%)	112 (50%)	112 (50%)
out_setup	8	0 (0%)	8 (100%)	0 (0%)
out_hold	8	8 (100%)	0 (0%)	0 (0%)
-----	-----	-----	-----	-----
All Checks	912	94 (10%)	258 (28%)	560 (61%)

Figure 7: Constraint analysis report with clock uncertainty removed.

With clock uncertainty removed, the number of timing violations is unchanged. Because the lab used very high clock frequencies (100-500 GHz), the timing violations that occurred were very large. Removing clock uncertainty reduced propagation delays by 0.5ps, which was not enough to remove any of the large timing delays in the design.